

THE SELF-HOSTED AI LEARNING PLATFORM ↘

LEARN THE FUTURE WAY.

An AI tutor fine-tuned on real Cambridge past papers — grounded in **official mark schemes**, running entirely on your own server. Zero third-party AI APIs. **Live and deployed today.**



Your data never leaves your server

LoRA fine-tune of Qwen2.5-0.5B runs in-process — no OpenAI, no vendor cloud.



Answers you can trust

Retrieval over 68k+ aligned questions returns the official mark scheme, attributed.



Snap & Solve vision

Photograph a diagram — the moondream2 VLM actually sees it, OCR fallback included.

BY THE NUMBERS

68k+

ALIGNED PAST-PAPER Q/A

\$0

PER-QUESTION AI COST

3

ROLE-BASED PORTALS

100%

SELF-HOSTED STACK

1

CRITERION 1 · 20 MARKS

INNOVATION,
NOVELTY & ORIGINALITY

EXCELLENT · 90-100%



Self-hosted, fine-tuned tutor — not a GPT wrapper

A LoRA fine-tune of Qwen2.5-0.5B on aligned Cambridge papers. Most edtech resells a vendor API; we own the model.



Mark-scheme-grounded answers

Retrieval returns the *official* Cambridge mark scheme, attributed — not a hallucinated guess.



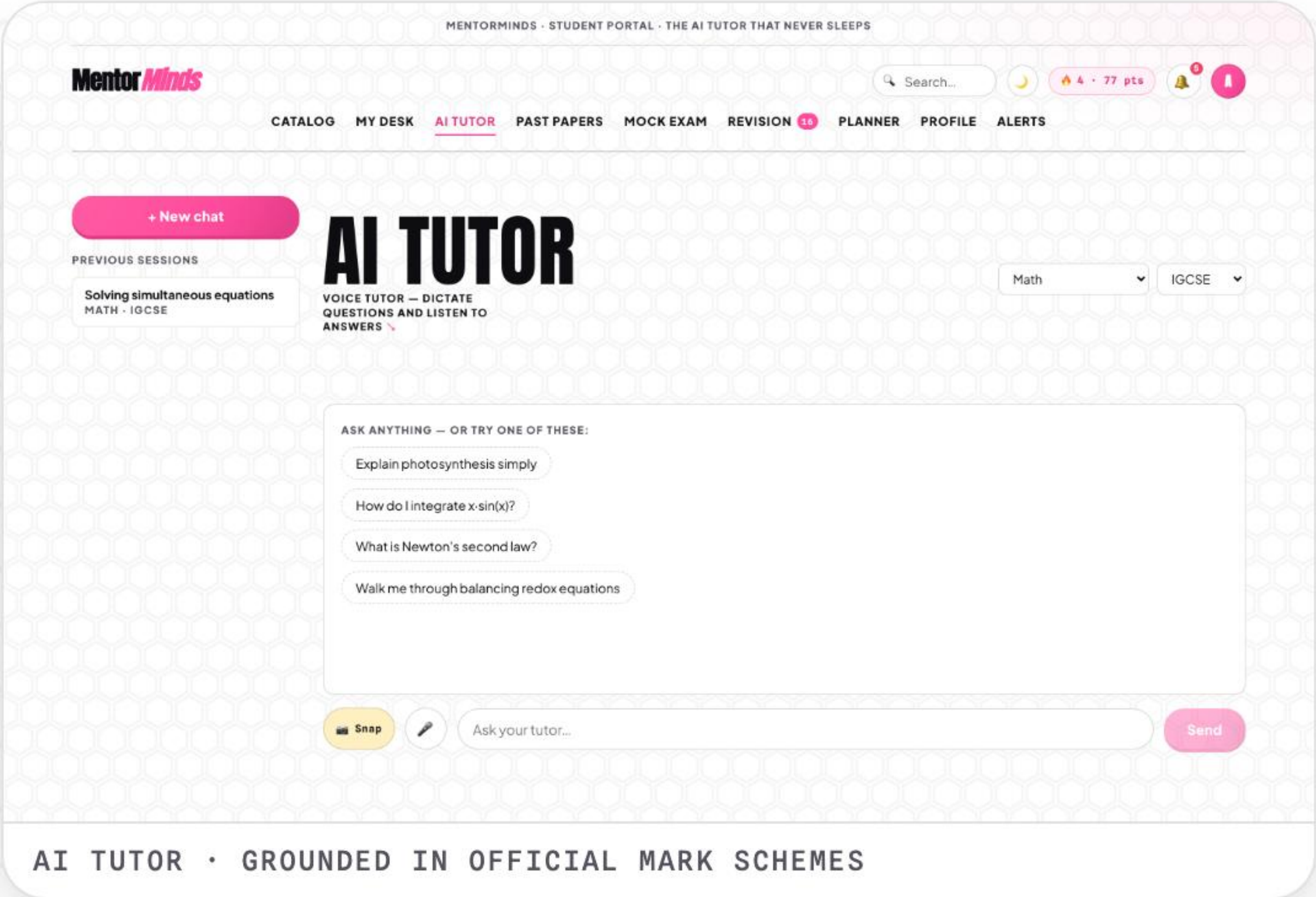
Snap & Solve VLM

moondream2 reads photographed diagrams/handwriting; clean text MCQs become auto-marked questions.



Adaptive learning loop

Weak-topic detection, spaced-repetition revision, and a dropout-risk model close the loop.





CRITERION 2 · 20 MARKS

TECHNICAL DEPTH, FEASIBILITY & STATUS

● EXCELLENT · NEAR PRODUCTION

3 Angular SPAs
STUDENT · STUDIO · ADMIN

▼ HTTPS · Caddy (auto-TLS) · Docker Compose · VPS

Django + DRF API
AUTH · COURSES · PLANNER

Channels WS
LIVE CHAT

Celery + Redis
JOBS · CACHE · QUOTAS

PostgreSQL
PRIMARY + REPLICA

▼ X-API-Key

FastAPI ml-service (HF Space)
QWEN2.5-0.5B + LORA · RAG OVER 68K+ Q/A

moondream2 VLM
/V1/VISION/ASK · OCR FALLBACK

OpenCV
PROCTORING · OMR · OCR

Status: live on VPS + HF Space

CI: tests + builds green

Dynamic-first: feature flags, live site settings

Sovereign: no third-party AI APIs

\$0 marginal inference cost



CRITERION 3 · 20 MARKS

IMPACT, CONTRIBUTION & SOCIAL VALUE

● EXCELLENT · SDG-ALIGNED



SDG 4 — Quality Education

Mark-scheme-grade tutoring for IGCSE/Cambridge students who can't afford private tutors.



\$0 per-question cost

A 0.5B model on cheap CPU means schools in emerging markets can run it sustainably.



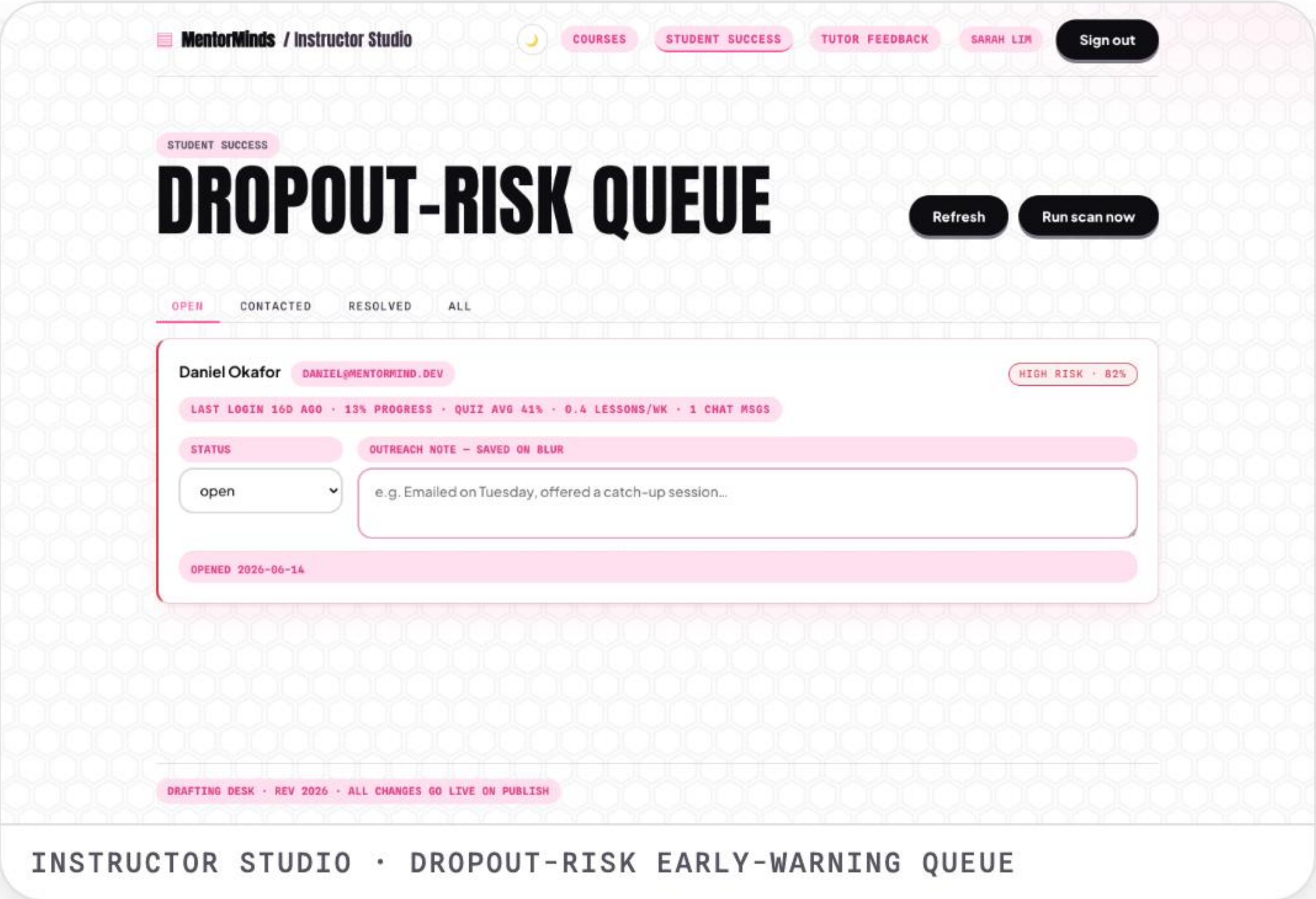
Data sovereignty

Schools self-host; student data never leaves their server — a real privacy guarantee.



Dropout-risk early warning

An ML model surfaces at-risk students so instructors can intervene before they quit.





CRITERION 4 · 20 MARKS

PRESENTATION, COMMUNICATION & VISUAL

● EXCELLENT · CLICKABLE DEMO



Cohesive EduNova design system

One bold, accessible visual language across all three portals.



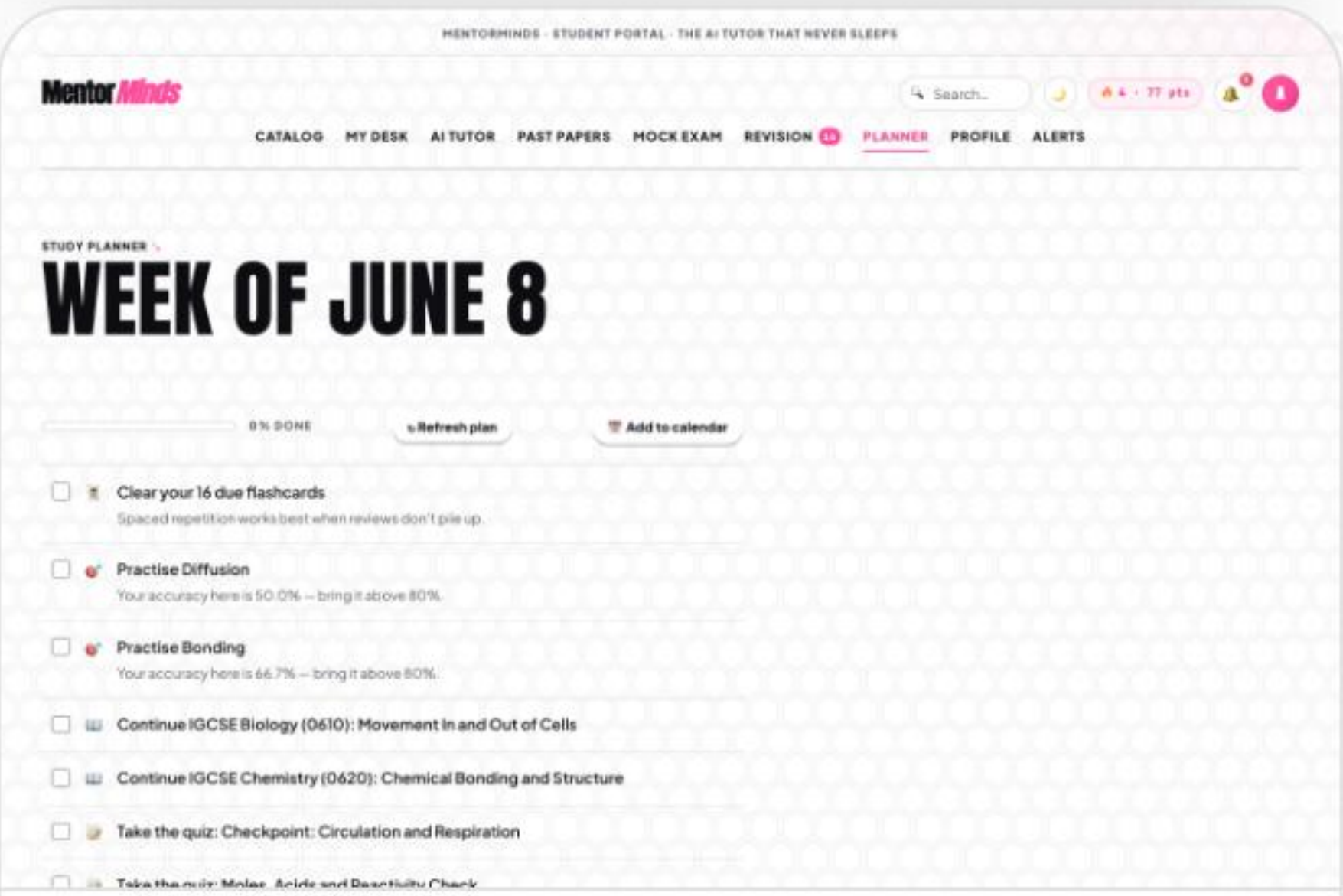
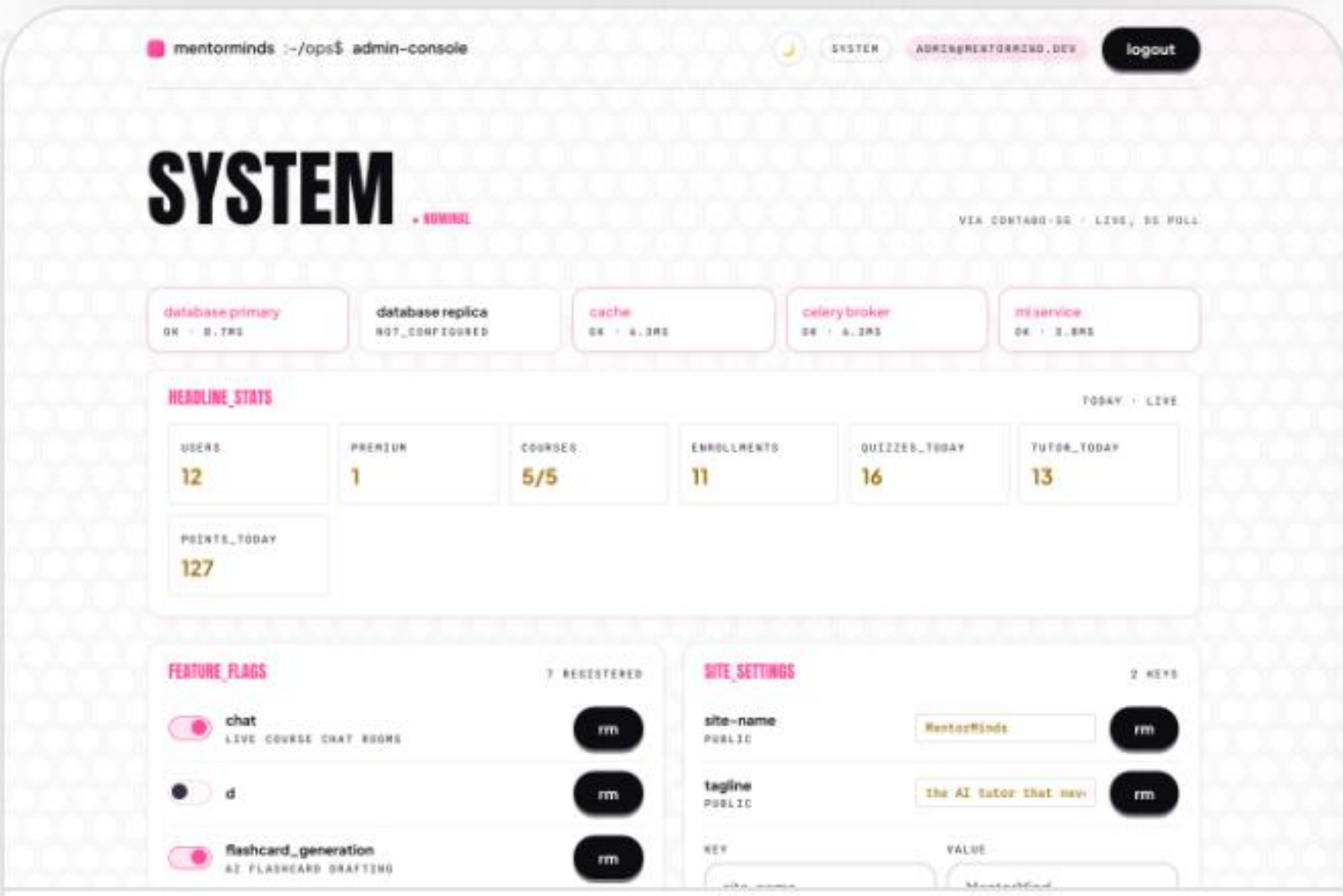
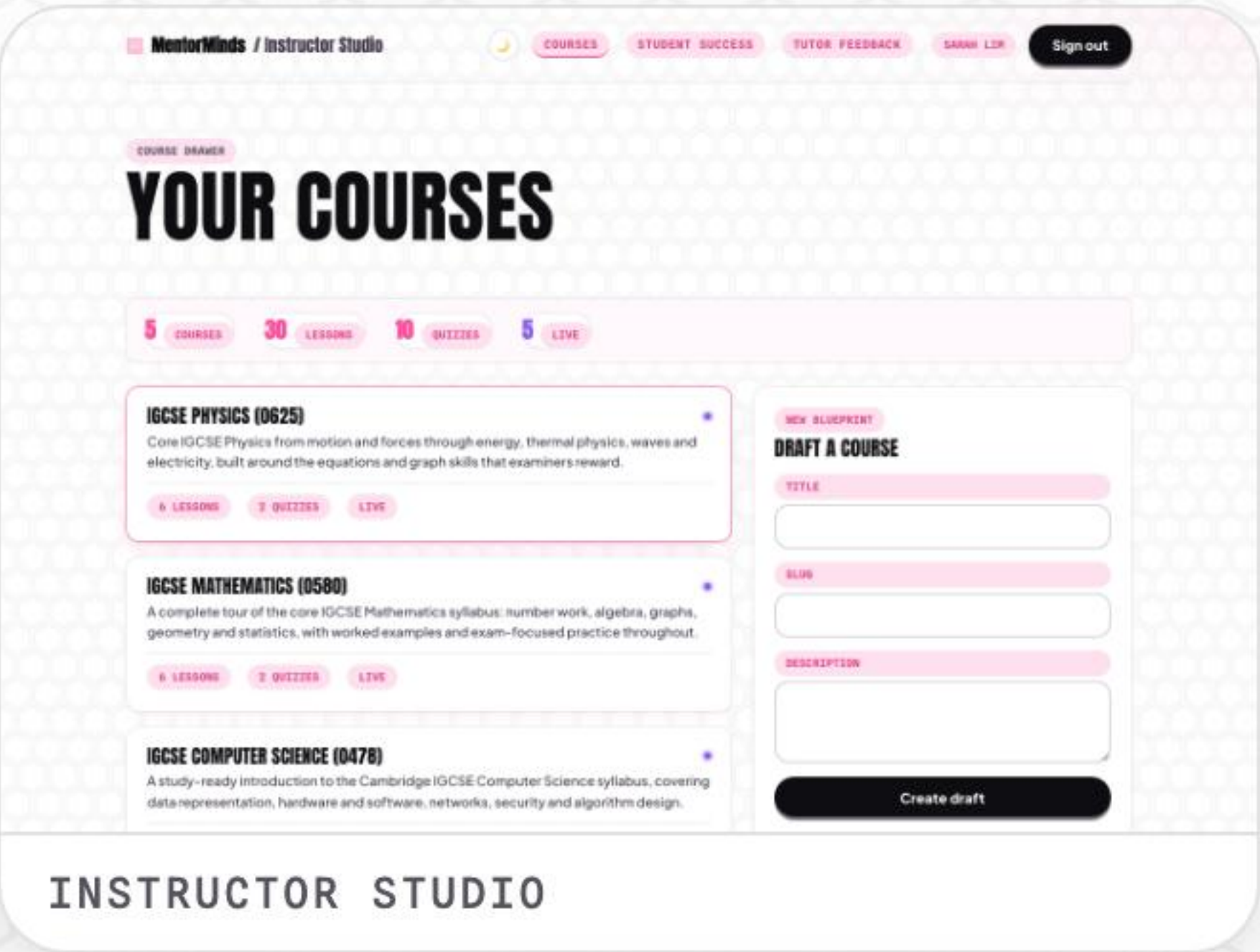
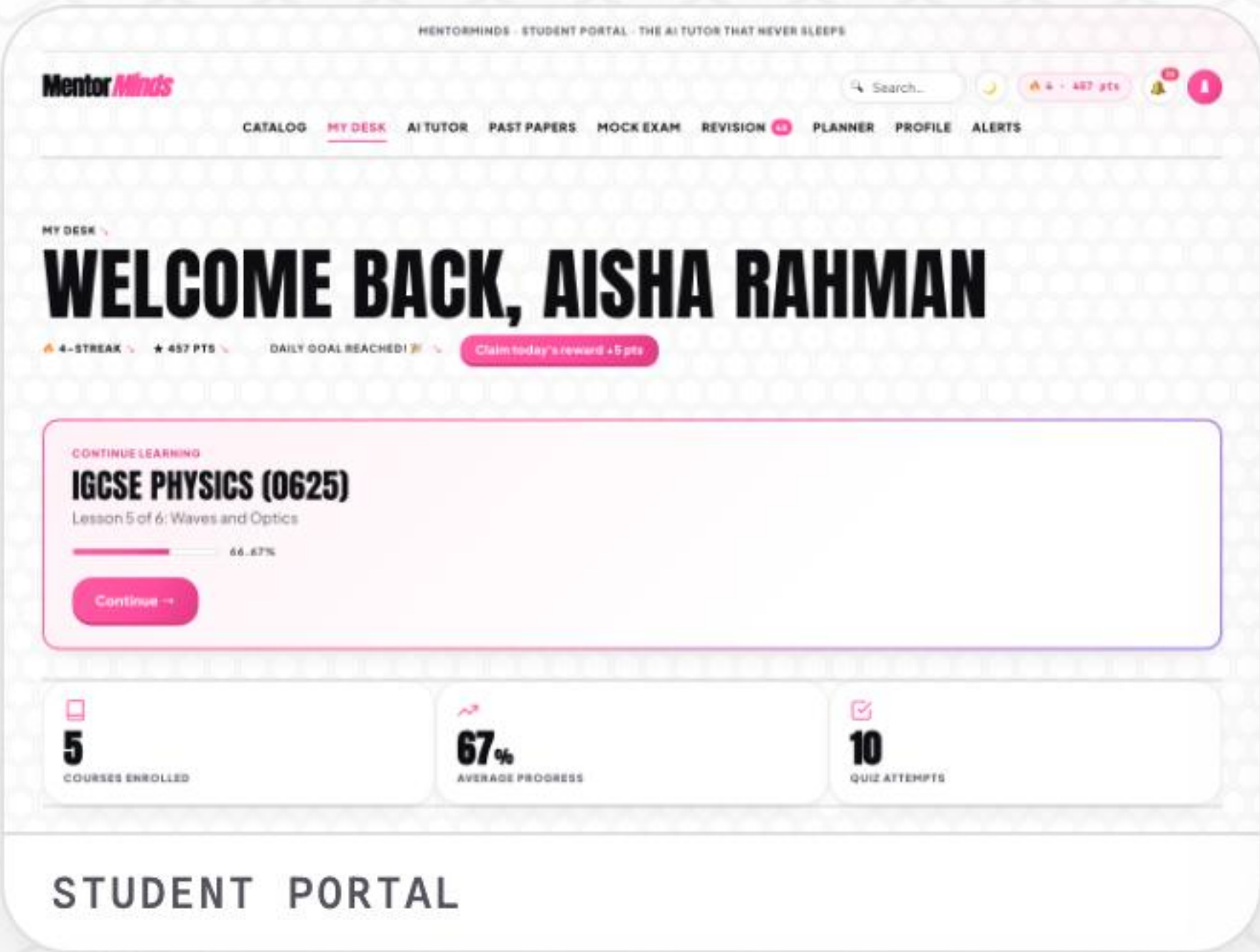
Judges can click the real thing

Not slideware — a live, seeded demo with login. QR + credentials on the next slide.



Responsive + Android build

Capacitor wrapper ships the student portal as a native app.



5

CRITERION 5 · 20 MARKS

COMMERCIALISATION & BUSINESS MODEL

● EXCELLENT · MODEL BUILT-IN



Target market

Schools & tutoring centres running Cambridge/IGCSE prep in price-sensitive regions.



Dual revenue model

Per-school self-host license (B2B) + premium student subscription (B2C) — the payments + premium flow is already in the product.



High-margin & scalable

\$0 marginal AI cost + one multi-tenant-ready stack = margin grows with each school added.



Defensible moat

Proprietary 68k+ aligned-paper corpus + a fine-tuned model competitors can't simply API-call.

UNIT ECONOMICS

\$0

COST PER AI ANSWER

B2B+B2C

REVENUE STREAMS

1

STACK, MANY TENANTS

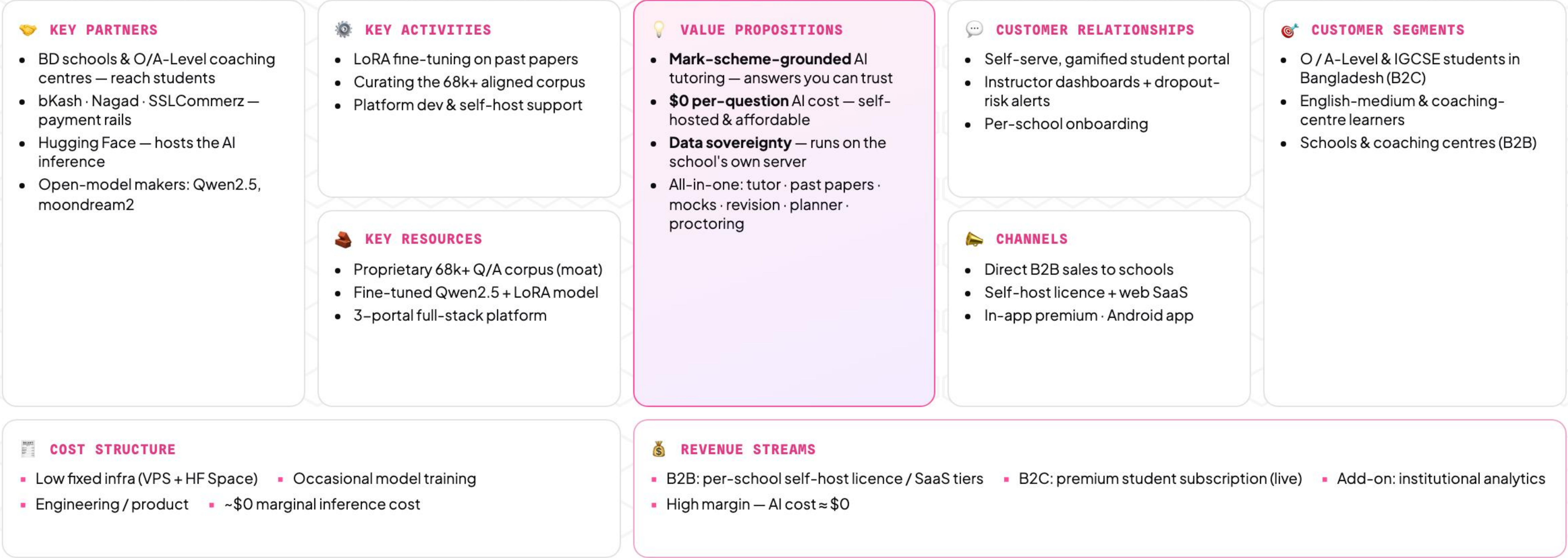
68k+

CORPUS MOAT

5

CRITERION 5 · SUPPORTING ARTEFACT

BUSINESS MODEL CANVAS



DON'T TAKE OUR WORD FOR IT ↘

CHECK EVERY CLAIM, LIVE.



mentorminds.duckdns.org

Student portal · login student@mentormind.dev / 12345678



/studio & /console

Instructor studio & admin ops console on the same domain.



ml-service API

arnob66666-mentormind-ml.hf.space — tutor, vision, grading endpoints.



Source + CI

github.com/Arnobrizwan/mentormind — green builds, tests, this deck.

